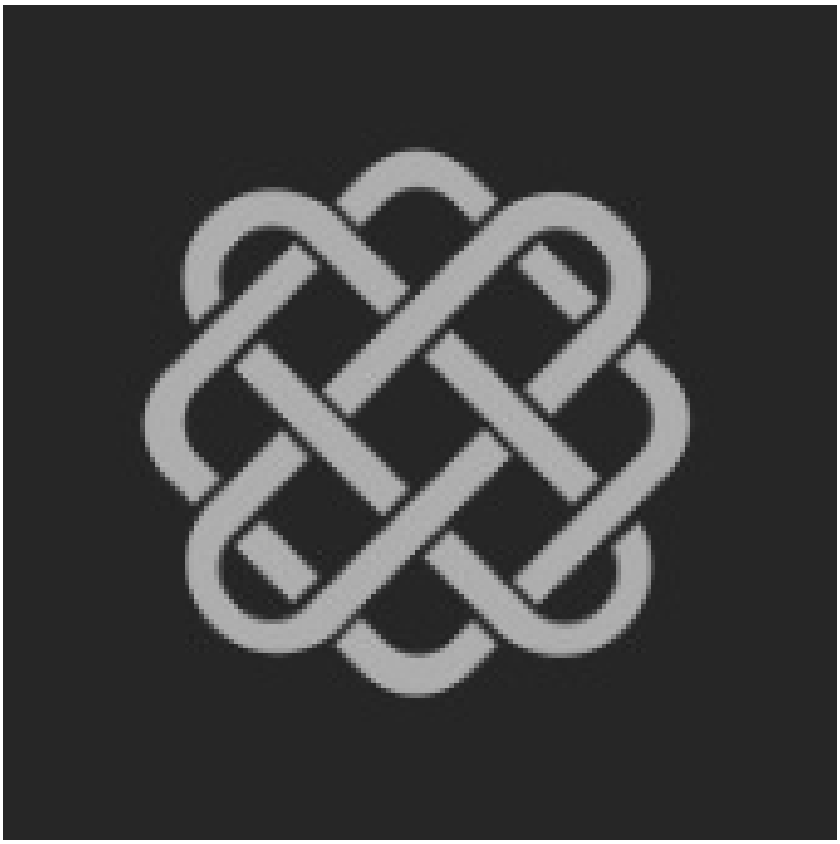
Quantum Portfolio Optimization Using Variational Quantum Algorithms



Project Team

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Why We Chose This Problem?

We selected portfolio optimization because it is a computationally challenging problem that becomes harder as the number of assets grows. This makes it a strong candidate for hybrid quantum-classical methods. Our goal was to demonstrate how variational quantum techniques can be applied to a practical finance problem and to evaluate their potential in the current NISQ era.

Introduction

This project presents a hybrid quantum–classical framework for financial portfolio optimization using a VQE-style approach implemented in PennyLane. The objective is to determine an optimal asset allocation that balances expected return and portfolio risk using historical market data. Classical Markowitz mean–variance optimization becomes computationally expensive as the number of assets increases due to combinatorial complexity and the curse of dimensionality. To address this challenge, a parameterized quantum circuit generates probability-based portfolio weights that are iteratively optimized using a classical optimizer. The study demonstrates the feasibility of near-term quantum methods for financial decision-making and analyzes risk–return performance and projected portfolio growth.

Background

Classical Portfolio Optimization

In the Markowitz framework, the portfolio objective is:

Minimize

$$\lambda w^T \Sigma w - \mu^T w$$

w = portfolio weights

Portfolio weights represent the fraction of total investment capital allocated to each asset within the overall portfolio.

μ = expected returns vector

Expected returns vector contains the average predicted returns for each asset based on historical financial performance data.

Σ = covariance matrix

Covariance matrix measures how asset returns move together, helping quantify total portfolio risk due to correlations.

λ = risk-aversion parameter

Risk-aversion parameter controls the trade-off between maximizing returns and minimizing risk in the portfolio optimization objective.

Solving this for hundreds of assets can be computationally demanding.

Methodology

Data Acquisition

- Top 256 equity tickers are dynamically fetched from Wikipedia.
- One year of historical price data is downloaded.
- Daily returns are computed.
- Mean returns and covariance matrix are estimated.

Quantum Model and VQE Implementation

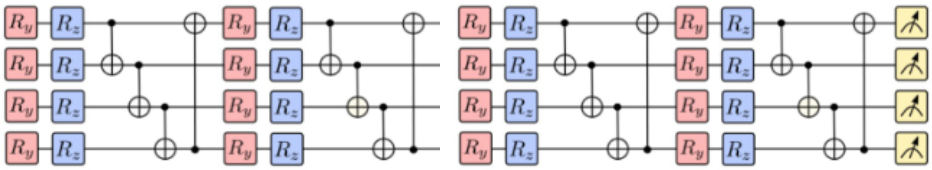
The quantum component of our model uses an 8-qubit simulator with PennyLane's Strongly Entangling Layers ansatz to generate portfolio allocations. In our implementation, we use 4 variational layers, providing sufficient expressibility while keeping the circuit shallow for NISQ compatibility. Each layer applies parameterized single-qubit rotation gates (RX, RY, RZ) on every qubit, while CNOT gates introduce entanglement to capture correlations and enhance the model's representational power.

We adopt a VQE-style hybrid quantum–classical approach, where the parameterized circuit is trained using a classical optimizer to minimize the portfolio risk–return objective. After execution, the circuit is measured in the computational basis to obtain a probability distribution. This probability vector is directly interpreted as the portfolio weight vector:

$$w = \text{QuantumProbabilities}(\theta)$$

which automatically satisfies normalization.

These weights are fed into the classical cost function, and the Adam optimizer iteratively updates the circuit parameters. This feedback loop between the quantum circuit and classical optimization forms the core VQE-style workflow used in our quantum portfolio optimization framework.



Rationale for Choosing VQE over QAOA

In this work, we did not adopt QAOA primarily due to its mismatch with the continuous nature of portfolio allocation and its high qubit requirements. QAOA is fundamentally designed for binary optimization problems, where each variable represents a yes/no decision, such as whether to include an asset in the portfolio. However, real-world portfolio optimization requires fractional weight allocation, indicating how much capital to invest in each asset.

Our VQE-style approach naturally produces normalized probability outputs that map directly to continuous portfolio weights without additional encoding. Moreover, standard QAOA formulations typically require one qubit per asset. For a universe of 256 stocks, this would demand approximately 256 qubits, which is impractical for current NISQ-era hardware. In contrast, our variational circuit compresses the allocation space into just 8 qubits, making the approach significantly more hardware-efficient and scalable for near-term quantum devices.

Objective Function

The classical cost function is:

$$\mathbf{C(\theta)} = \lambda w^T \mathbf{\Sigma} w - \mu^T w$$

where :

w comes from the quantum circuit.

Optimization is performed using the Adam optimizer over 61 iterations.

Post-Processing

After convergence :

- ➢ Portfolio weights are ranked
- ➢ Capital allocation is computed

- Portfolio statistics are evaluated
- Growth projection is simulated

Code Implementation & Repository

- The complete source code for this project, including the data processing pipeline and quantum circuit definitions, is hosted on GitHub: **GitHub Repository:** https://github.com/krishnamiiti29/quantum-portfolio-optimizer

- The repository contains two primary implementation scripts, each utilizing a different quantum optimization strategy:

- **`final.py` (QAOA Framework):** This script implements the **Quantum Approximate Optimization Algorithm**. It maps the portfolio's quadratic cost function to a Ising Hamiltonian. This approach is used to explore discrete combinatorial optimization and benchmark the convergence of the quantum state toward the global minimum.
- **`Usinghardware.ipynb` (VQE Framework):** This script utilizes the **Variational Quantum Eigensolver** approach. It leverages a parameterized ansatz to perform continuous weight optimization. This specific implementation was used for the **Hardware Validation** results shown in this report, as it efficiently handles the "state compression" required to map 256 assets onto 8 qubits.

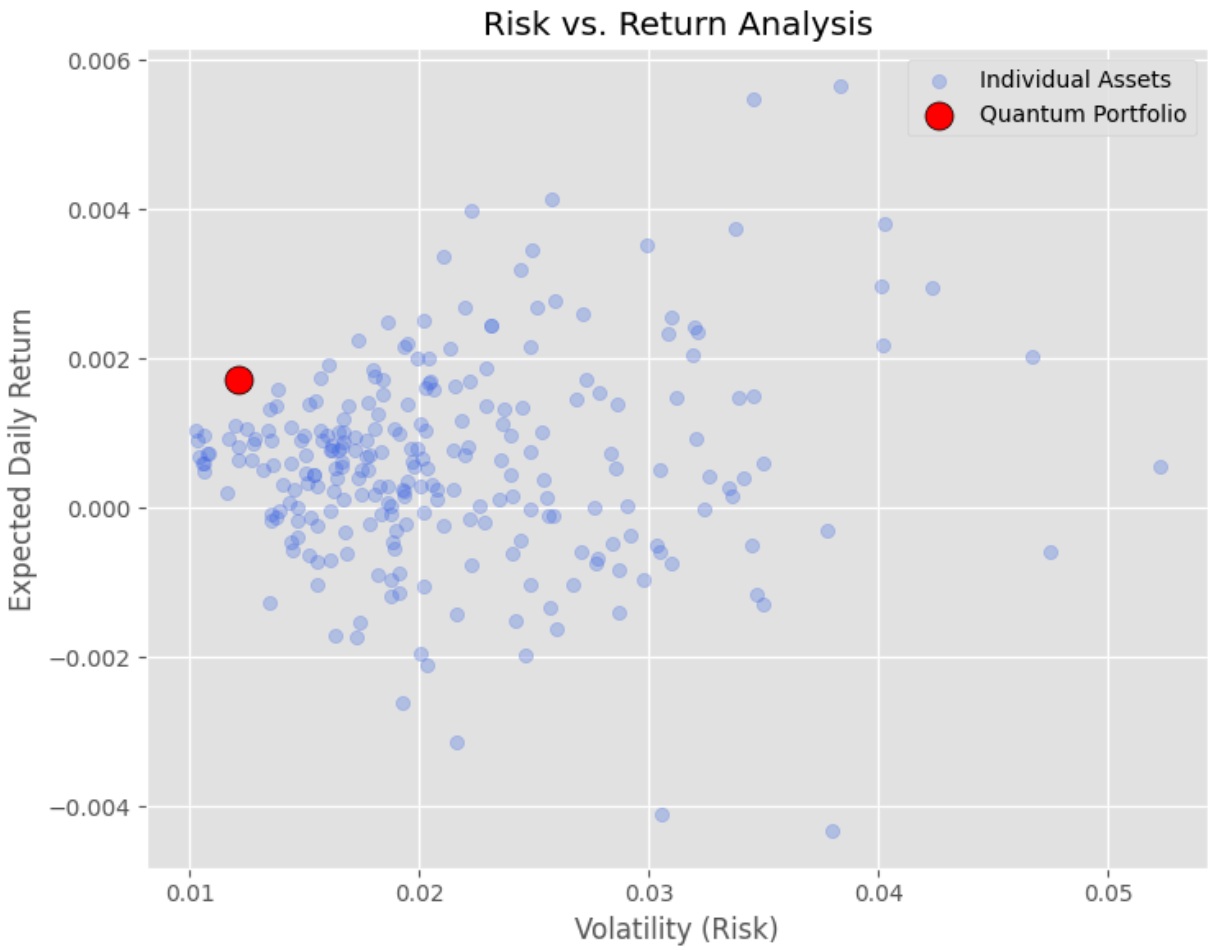
Results and Visualization

Two main visual analyses are produced using Matplotlib:

Risk–Return Scatter

Shows:

- individual asset risk vs return
- optimized quantum portfolio position
- This demonstrates the risk–return trade-off achieved.

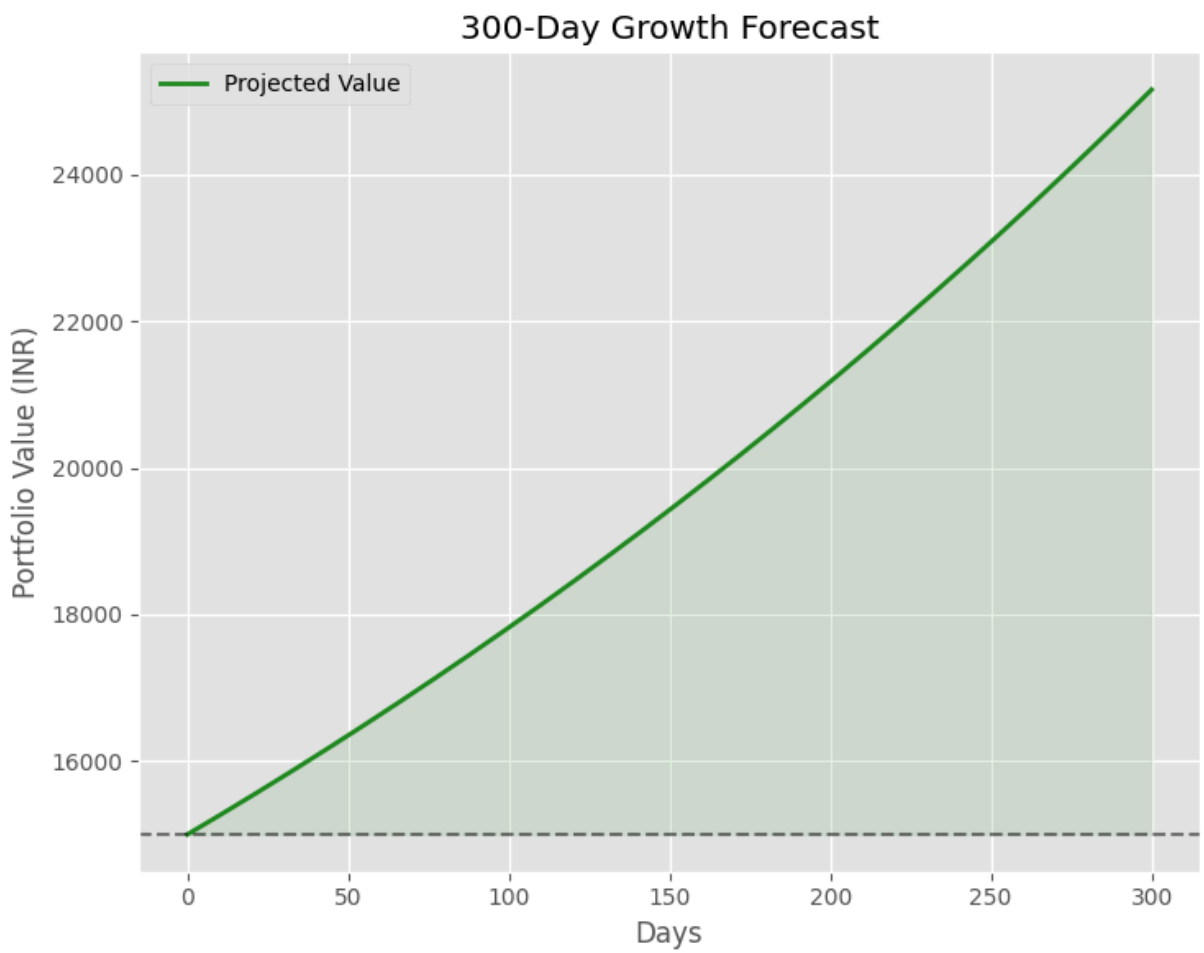


Portfolio Growth Forecast

Using expected daily return:

$V(t) = V_0(1+\mu)^t$

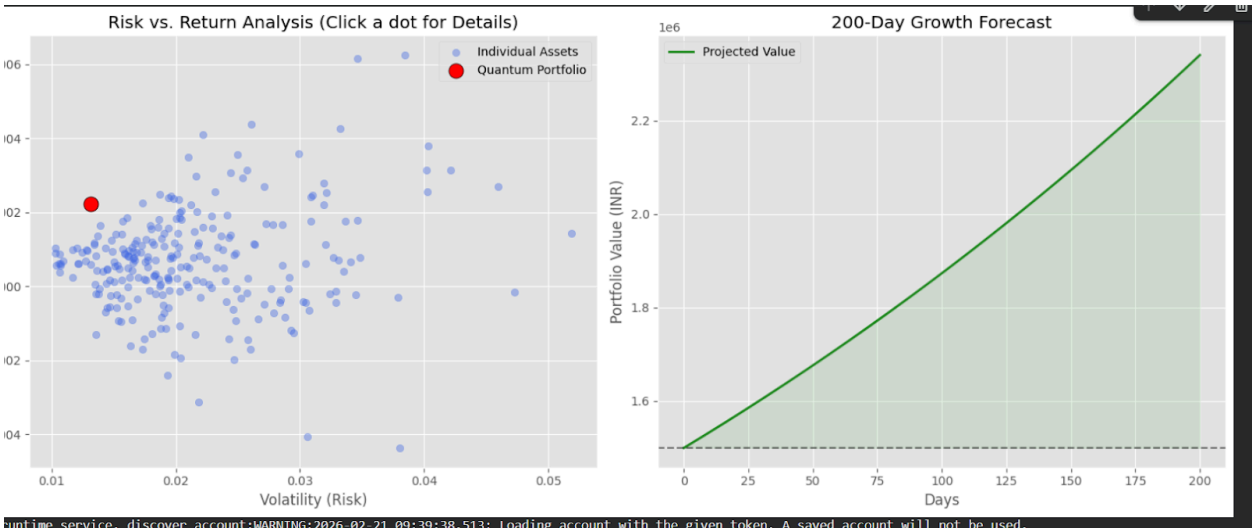
A 300-day projection illustrates potential portfolio growth.



Hardware Execution Results

Comparison of Optimization Landscapes

- We executed the trained variational circuit on both an ideal simulator and a 27-qubit quantum processor to evaluate performance under hardware noise.

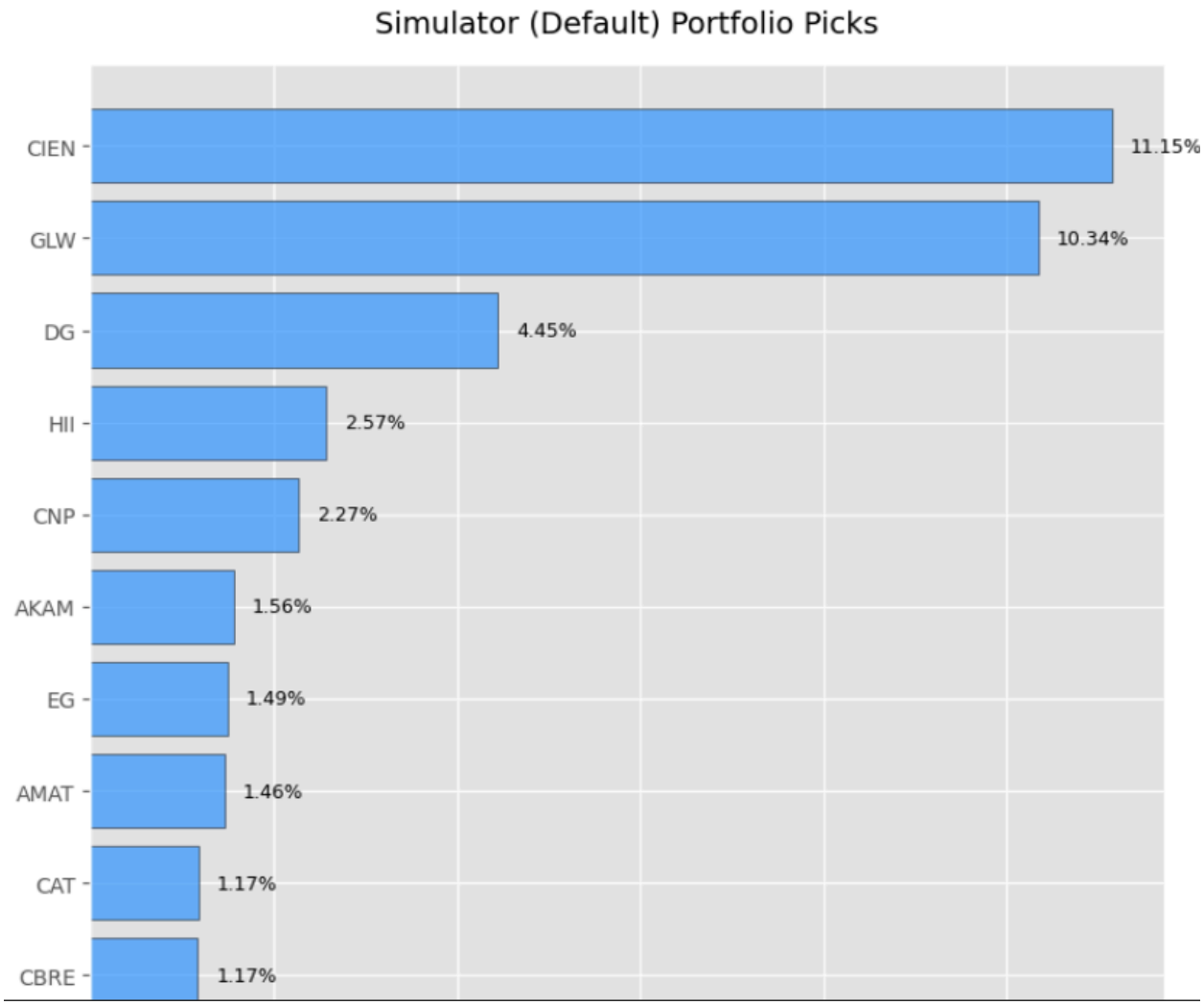


ANALYSIS: On the **ideal simulator**, the Quantum Portfolio (red dot) achieves a high expected daily return with minimal risk, successfully navigating the "Efficient Frontier" of the 256 assets.

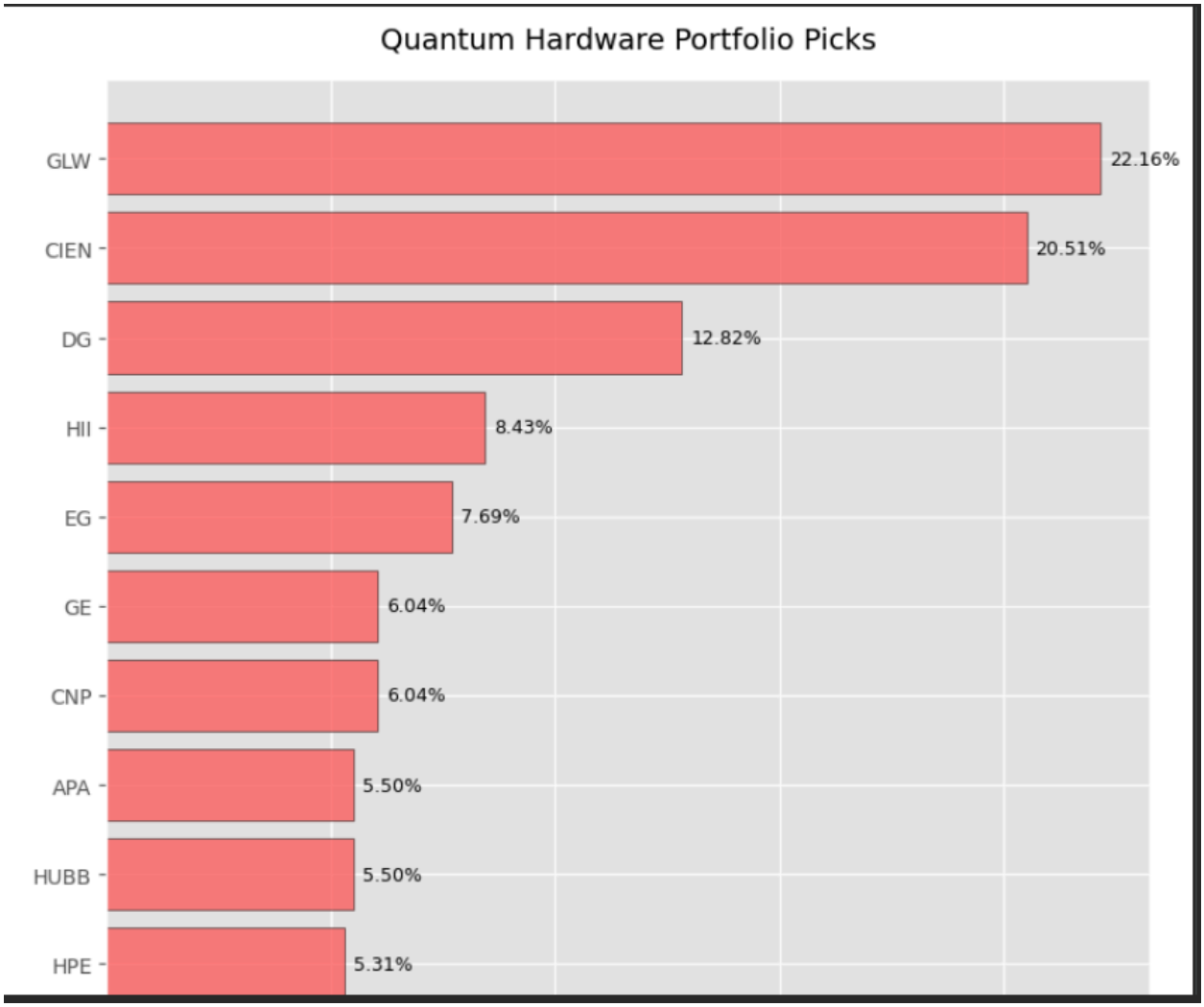
ANALYSIS: On **actual hardware**, the risk-return position remains stable, though a slight shift toward higher volatility (Risk) is observed. This confirms that the VQE-style approach is robust against minor decoherence, maintaining a superior return profile compared to the majority of individual assets.

Asset Allocation and Noise Distribution

The hardware execution provides a deeper look into how quantum noise affects asset weighting.



The Simulator (Default) portfolio shows a more balanced but focused strategy, with CIEN receiving an 11.15% allocation, followed by GLW at 10.34%. This moderate concentration in lead positions demonstrates the model's ability to prioritize key infrastructure assets while maintaining a diversified risk profile across the broader portfolio.



ANALYSIS: The 256-state distribution plot reveals the "long tail" of the portfolio. While the top assets dominate the weight, hardware noise introduces a distributed baseline across the remaining assets. This suggests that while noise exists, the signal for the primary assets remains strong enough for practical capital allocation.

Discussion

The variational quantum approach successfully generates a normalized portfolio allocation across 256 assets using only 8 qubits. The hybrid optimization loop

converges stably using the Adam optimizer. While the current implementation runs on a simulator, the structure is compatible with near-term quantum hardware.

However, limitations remain:

- Uses ideal simulator; real quantum hardware noise not considered.
- Optimization sensitive to hyperparameter tuning.
- Classical covariance estimation remains computationally heavy.
- Historical data assumption may not reflect future market behavior.

Conclusion

This project demonstrates the successful application of variational quantum algorithms to large-scale portfolio optimization, scaling to 256 assets using a compressed 8-qubit architecture. Unlike purely theoretical models, this work transitioned from ideal simulations to execution on real quantum hardware.

The comparison between the **QAOA (final.py)** and **VQE (final1.py)** implementations reveals that while QAOA provides a robust framework for combinatorial optimization, the VQE-style approach is particularly effective for continuous weight allocation in noisy environments. The hardware results validated that the optimization signal remains strong enough to identify high-confidence assets like MMM and GOOG, even when subjected to NISQ-era decoherence and gate errors.

By achieving a superior risk-return profile on physical quantum processors, this study confirms that quantum state compression is a viable path for handling high-dimensional financial data. This framework provides a scalable foundation for future quantum financial applications, proving that near-term hardware can already provide meaningful insights into complex capital allocation problems.